

## International trade

The buying and selling of goods and services across national borders is known as international trade. International trade is the backbone of our modern, commercial world, as producers in various nations try to profit from an expanded market, rather than be limited to selling within their own borders. International trading is nothing new to the business world. We have been trading across boundaries ever since we found ways to move past borders in the latest modes of transportations but the way trading is done these days is far more complicated and lucrative than it used to be. Industrialization, globalization and formation of many multinational corporations have all changed the way nations deal with each other. There are many reasons that trade across national borders occurs, including lower production costs in one region versus another, specialized industries, lack of surplus of natural resources and consumer tastes.

One of the most controversial components of international trade today is the lower production costs of developing nations. One of the largest movers in the international trading world than we have today is China where labor is plentiful and cheap. Many labor intensive products designed and produced by United States and other European countries are assembled or manufactured in China where labor is inexpensive. This is typical because it's a move that can save the original country a lot of time and money. Furthermore, with the opening of door of China, citizens now have more income opportunities to make life better.

There is currently a great deal of concern over jobs being taken away from the United States, member countries of the European Union and other developed nations as countries such as China, Korea, India, Indonesia and other produce goods and services at much lower costs. Both the United States and the European Union have imposed severe restrictions on imports from Asian nations to try to stem this tide. Clearly, a company that can pay its workers the equivalent of dollars a day, as compared to dollars an hour, has a distinct selling advantage. Nevertheless, American and European consumers are only too happy to lower their costs of living by taking advantage of cheaper, imported goods.

Even though many consumers prefer to buy less expensive goods, some international trade is fostered by a specialized industry that has developed due to national talent and/or tradition. Swiss watches, for example, will never be price competitive with mass produced watches from Asia. German cutlery, Scottish wool, fine French silks and other such products always find their way onto the international trade scene.

**Difference between domestic and foreign trade:** Domestic trade is the buying and selling of goods and services within the boundaries of a country while foreign trade is the buying

## International trade

and selling of goods and services across the boundaries i.e. between countries. There are many differences between domestic and international trade as follows:

1. **Distance**: The distance involved in export of goods in external trade is generally greater than on the domestic trade
2. **Language differences**: There are differences in the languages of the nations of the world. The overseas traders should be very careful in preparing the publicity material in the languages of the trading country
3. **Cultural difference**: A producer should have full knowledge about the market of his products. For exporting goods particularly a thorough research is undertaken.
4. **Technical difference**: In the national market the difference in the technical specification for goods and their requirements is not wide.
5. **Tariff barriers**: In the national trade, there are no custom duties, exchange restrictions, fixed quotas or other tariff barriers.
6. **Documentations**: In the home trade there are few documents involved in the exchange of goods.
7. **Payments**: In the internal trade, the goods are exchanged in the currency unit of the country. In case of foreign trade currencies differ widely throughout the world and those also vary in value.
8. **Transport and insurance cost**: The transport and insurance costs are less in case of domestic trade. For the exports, on the other hand the cost of transport is high and the insurance is complicated.

**Benefits of international trade:** Each country has a different allocation of natural and human resources i.e. factor endowment. A given country will be well suited to the production of some goods (or the provision of certain services) and totally unsuited to the production of other. Particularly in the field of agriculture, the differences in climate can be immense; it is simply impossible to grow certain agriculture products in certain countries.

Thus international trade is mostly restricted to trade in goods and services, and only to a lesser extent to trade in capital, labor or other factors of production as they are typically more mobile within a country than across countries. Then trade in goods and services can serve as a substitute for trade in factors of production. Instead of importing the factor of production a country can import that make intensive use of factors of production and are thus embodying the respective factor. An example is the import of labor intensive

## International trade

goods by the United States from China. Instead of importing Chinese labor the United States is importing goods from China that were produced with Chinese labor.

International trade enables a nation to specialize in those goods it can produce most cheaply and efficiently. Trade also enables a country to consume more than it would be able to produce if it depended only on its own resources. It enlarges the potential market for the goods of a particular economy. Trade has always been the major force behind the economic relations among nations.

International trade is a major source of revenue for many countries which exports goods of greater value than their imports. This revenue then is used for economic growth and development.

The more open to foreign trade a country is the more competition the home producers will face and the more efficient they will have to be. This is not only good in an economic sense (greater efficiency) but domestic consumers enjoy lower prices and, therefore, increased consumer surplus.

International trade gives the firms a large platform to introduce their goods and services. They have access to a larger market enabling them to take a greater advantage of economies of scale. They will be able to purchase raw materials and other component parts more easily and more cheaply. It is not only the producers that benefit but also the consumers. They can purchase goods from other countries in their local markets. They have more variety to choose from. Because of competition each consumer will benefit from low priced better quality products. International trade increases world output.

The important concept here to note is regarding free trade i.e. no tools of protectionism. This allows countries to specialize in the products in which they have comparative advantage. Comparative advantage occurs when a country has lower opportunity cost than other. It then specializes internationally and exports that product.

**Gains from international trade:** In order to calculate gains from international trade we have to assume a very simple model of the world economy as follows:

1. Two countries, two goods - the theory is no different for larger numbers of countries and goods, but the principles are clear and the argument easier to follow in this simpler case. The two countries are country A and country B producing cars and wheat.

## International trade

2. Full employment - each country is on rather than within its PPF. We assume, therefore, that each country is using all of their resources and in the most efficient way.
3. Constant opportunity costs - the PPFs are straight lines. This means we are assuming that all resources used are equally good at making cars and wheat. The straight line PPF means that we are assuming constant opportunity costs of producing cars and wheat in terms of each other.
4. Negligible transport cost - cost is not a cause of concern when countries decide to trade. It is ignored and not factored in.

**Concept of Absolute advantage:** According to the principle of absolute advantage, a country has an absolute advantage over its trading partner if it is able to produce more of a good or service with the same amount of resources. To identify absolute advantage, you must identify the country which can produce more of a good than other country.

**Concept of comparative advantage:** To identify which country has the comparative advantage, you will be required to identify the opportunity cost ratios. A country having comparative advantage will have lower opportunity cost than other country.

Several cases will clarify the concepts -

### Example 1

|           | Good X | Good Y |
|-----------|--------|--------|
| Country A | 100    | 400    |
| Country B | 200    | 600    |

We are taking 2 countries and 2 goods because we are taking assumption of having bilateral trade.

To identify the absolute advantage, we have to see which country can produce goods more than the other. Country A can produce 100 units of X and 400 Units of Y, while country B can produce 200 units of X and 600 units of Y, so clearly country B has the absolute advantage in both the goods but that does not mean that it is efficient in producing both the goods.

To identify the comparative advantage, we are required to calculate the opportunity cost ratios

## International trade

|               |           |           |
|---------------|-----------|-----------|
| For country A | X : Y     | Y : X     |
|               | 100 : 400 | 400 : 100 |
|               | 1 : 4     | 1 : 0.25  |
| For country B | X : Y     | Y : X     |
|               | 200 : 600 | 600 : 200 |
|               | 1 : 3     | 1 : 0.33  |

You will always be required to follow these steps. When comparing X : Y for country A and B, country B is more efficient as it possess the comparative advantage which is lower opportunity cost i.e. it has to forgone less number of good Y to produce good X then country A. When you have identified that a country possess a comparative advantage in one goods for e.g. in good X, it's obvious that other country will have comparative advantage in other good, in this case country A has comparative advantage in good Y, which can also be proven through the calculation as it has lower opportunity cost in producing goods Y. countries which has comparative advantage in a particular good like country A in good Y and country B in good X will specialize in that product and then exports that product, one country's export is other country's import.

### Example 2

|           | Good X | Good Y |
|-----------|--------|--------|
| Country A | 100    | 300    |
| Country B | 200    | 400    |

Solve the above example; follow the steps as performed in earlier example.

### Example 3 - What if the opportunity cost ratio are same?

|           | Good X | Good Y |
|-----------|--------|--------|
| Country A | 100    | 200    |
| Country B | 200    | 400    |

Absolute advantage is with country B

When calculating comparative advantage:

|           |         |           |
|-----------|---------|-----------|
| Country A | 1X : 2Y | 1Y : 0.5X |
| Country B | 1X : 2Y | 1Y : 0.5X |

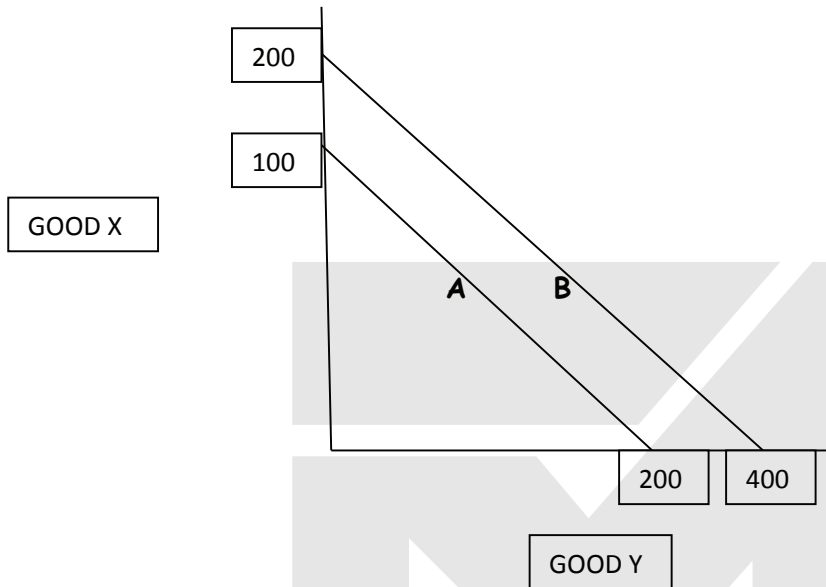
As we can see in the above calculation, the opportunity cost ratios are same, this will result in no gains from trade i.e. trade will not take place.

- Cease the trade
- Trade will not take place

## International trade

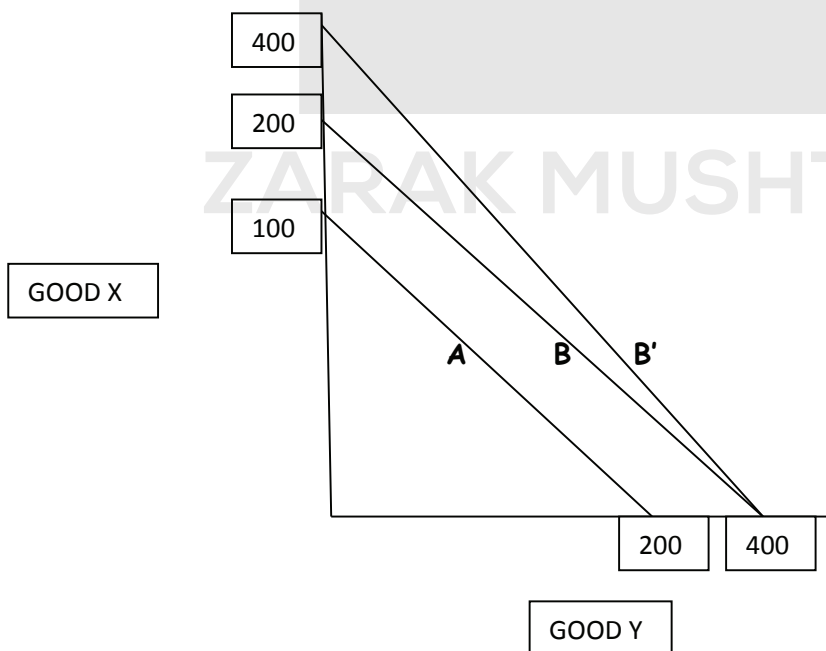
- There will be no gains from trade

This represents that production possibility curve (Linear) will be parallel to each other.



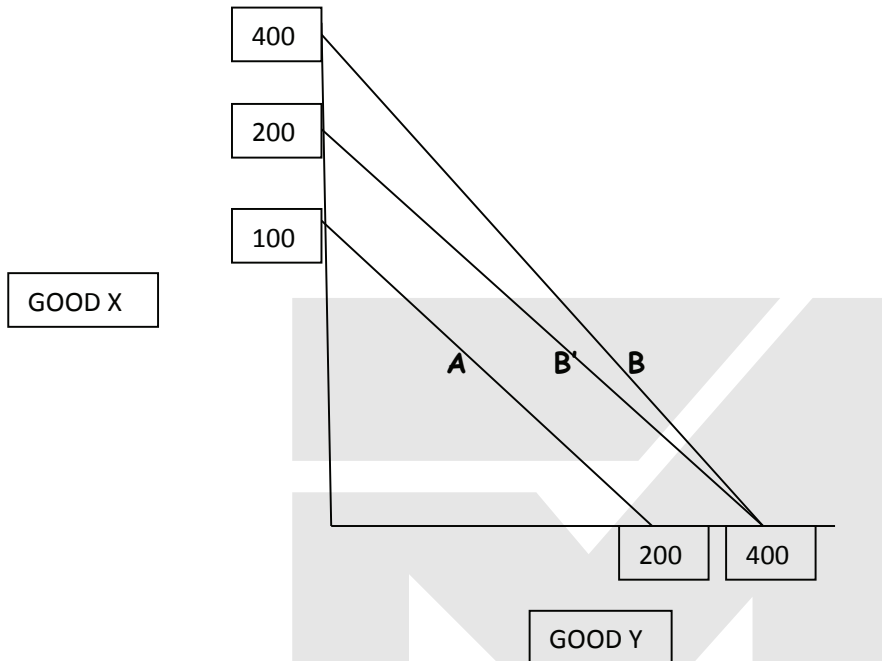
Mutually beneficial trade will take place if the domestic opportunity cost ratios are different. But the trade will not take place if the domestic opportunity cost ratios are similar.

**Example 4 - Earlier it was A and B, after it was A and B'. Solve the case.**



## International trade

**Example 5** - Earlier it was A and B', after it was A and B. Solve the case.



### Example 6

Country X    25 tonnes of maize or 5 tractors

Country Y    25 tonnes of maize or 5 tractors

Country Z    25 tonnes of maize or 5 tractors

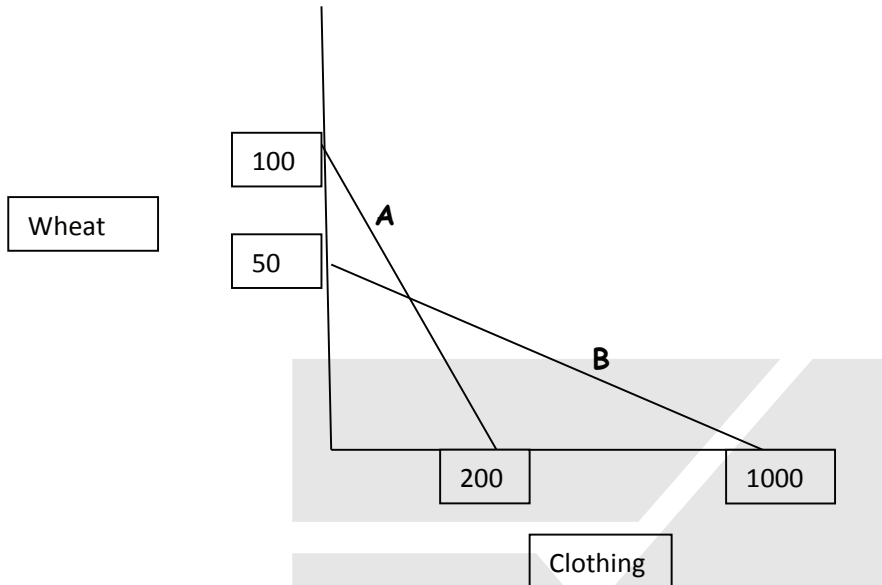
Solve the above example

### Example 7 - Find absolute and comparative advantage

|           | Output of X | Output of Y |
|-----------|-------------|-------------|
| Country 1 | 70          | 30          |
| Country 2 | 50          | 25          |

## International trade

### Example 8 - Find absolute and comparative advantage



**Remember -** If transport cost per unit of each commodity is equal to exchange rate, this would exactly eliminate the benefits of trade and countries even do not choose to trade having comparative advantage, if transport cost is higher than opportunity cost ratios.

### Example 5

The table shows the number of goods X and Y which two countries produce. Each country uses half of its resources to make each good

|         | Country 1 | Country 2 |
|---------|-----------|-----------|
| Good X  | 100       | 300       |
| Goods Y | 200       | 400       |

Later each specializes in the product in which it has a comparative advantage. Which rate of exchange would be suitable so each country gains from trade?

\*Remember, rate of exchange must lie between the opportunity cost ratios of two country, then only they will gain from trade.

A  $1X = 1Y$

B  $1X = 1.5Y$

## International trade

C  $1X = 2Y$

D  $1X = 3Y$

### Trading possibility curve

A trading possibility curve shows how an economy can benefit from specializing and trading. For example, Figure 25.4 shows a country that is able to produce a maximum of 50 million units of clothing or 100 million units of food. For every unit of clothing it produces it gives up the opportunity to produce two units of food. Engaging in international trade may enable the country to specialise in clothing production and import three units of food for every unit of clothing it exports.

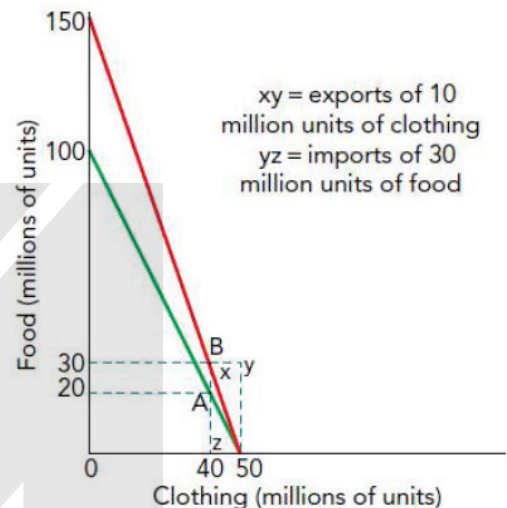


Figure 25.4: Trading possibility curve

In this case, trading will increase the total quantity of products the country can consume. If the country had initially produced 40 million units of clothing, the maximum output of food it could produce would be 20 million units. However, if it engages in international trade, it can specialise in clothing production, making 50 million units of clothing. It can export ten million units of clothing in exchange for 30 million units of imported food, enabling it to move from point A to point B and enjoy more products.

### PROTECTIONISM

**Protection;** refers to prevention of market forces working freely

**Protectionism** represents any attempt by a government to impose restrictions on trade in goods and services between countries. These restrictions are basically imposed to protect the domestic economy from foreign competition. There are different types of restrictions which a government can impose. Some of them are as follows:

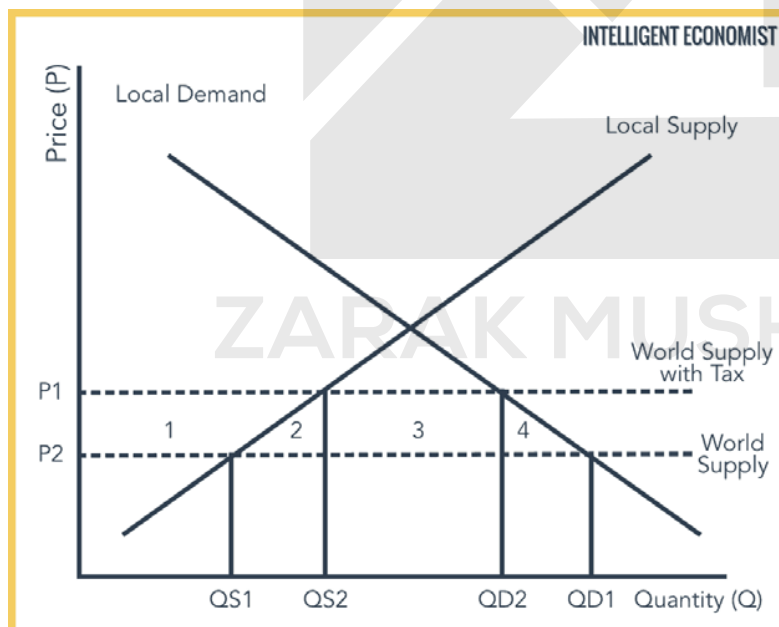
- Tariffs - import taxes
- quotas - quantitative limits on the levels of imports allowed
- voluntary export restraint arrangements - where two countries make an agreement to limit the volume of their exports to one another over an agreed period of time

## International trade

- Embargoes - a total ban on imported goods.
- Intellectual property laws (patents and copy rights)
- Export subsidies - a payment to encourage domestic production by lowering their cost
- Exchange controls - limiting the amount of foreign exchange than can move between countries.

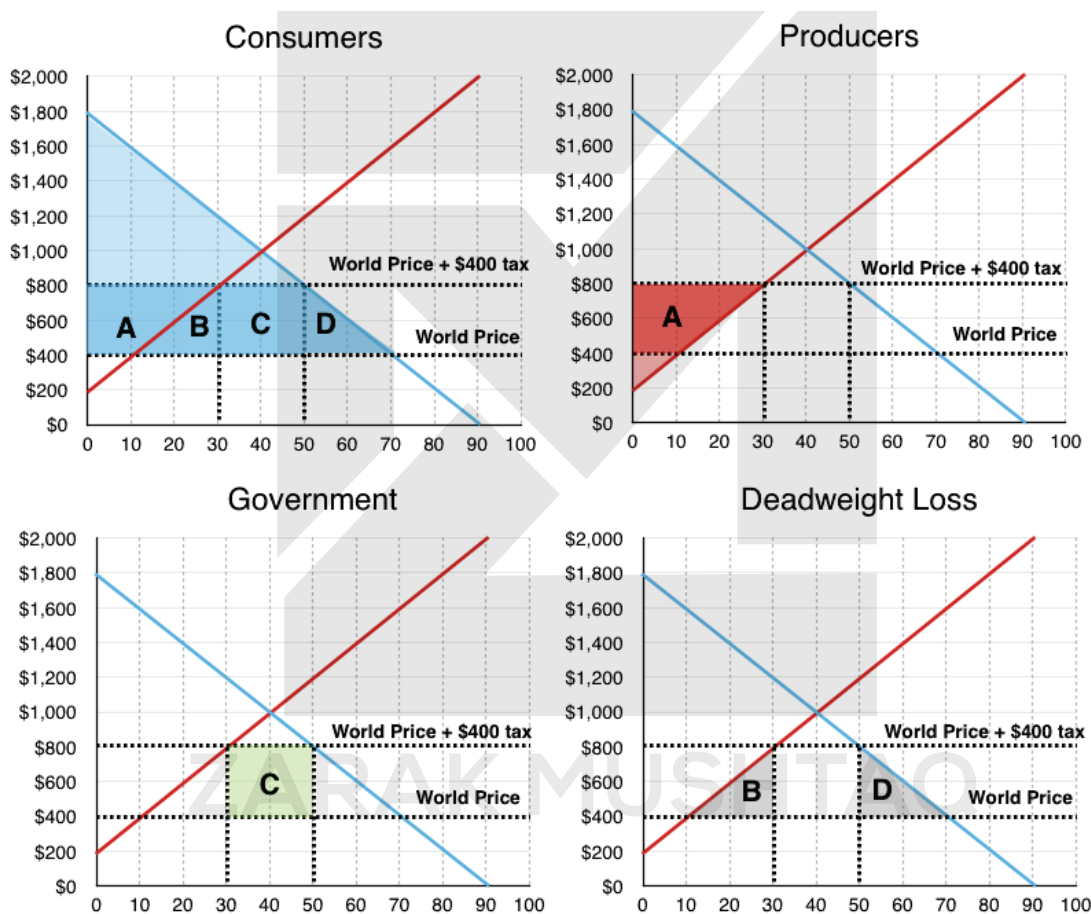
Quotas, embargoes, export subsidies and exchange controls are all examples of non-tariff barriers to international trade.

A tariff is, quite simply, a tax levied by the government on imported goods. This forces the price of the good up and so makes it relatively expensive in the home market. This benefit the domestic industry as its product now appears relatively cheaper to home consumers. The home industry has, in effect, been protected from the foreign competition. Tariff is one of the major sources of revenue for the government; it is imposed in two ways i.e. specific tariff and advalorem tariff. Specific tax is a fixed amount of tax per unit of the commodity imported. Advalorem tariff is the percentage of the prices or value of the commodity.



## International trade

In the figure there is a perfectly elastic worked supply which shows that an individual importing country cannot affect the world market price by making changes in the quantity of imports. At the world price  $p_2$ , the domestic producers are supplying  $Q_1$  quantity but the demand is  $Q_4$  and this excess demand of  $Q_4 - Q_1$  will be met through imports. By imposing tariff, the world price increased to  $p_1$  at which the domestic producers increased the quantity supplied to  $Q_2$  but the domestic demand is  $Q_3$  which is met through imports. Tariffs therefore reduced imports from  $Q_4 - Q_1$  to  $Q_3 - Q_2$  and consumer surplus reduced by the areas 1 + 2 + 3 + 4.



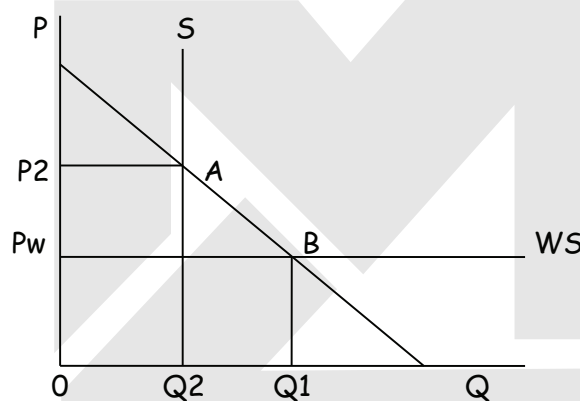
Tariff is a tax on imported goods; it increases revenue for the government when the demand for imported goods is inelastic. Government will be successful in achieving its target of reducing imports, if demand for imports will be elastic i.e. a small change in price will bring a larger percentage change in quantity demand, so it all depends on price elasticity of demand.

## Non-tariff barriers

## International trade

Due to the fact that the tariff has always been the main method used for protection (although that is not so true nowadays), economists often lump up all the other methods of protection into one group called non-tariff barriers. Here are some of the factors of the other forms of protection.

1. **Quotas:** A quota is a physical limit on the number of units of the product that can be imported into a country. The restriction in supply has similar effect to the tariff in that the price will rise, demand will fall and there will be a welfare loss to society. The difference is that with the tariff the domestic government receives some tariff revenue, with the quota; the benefit of the increased price goes to the foreign firms who provide the imports. Again the effectiveness will depend on price elasticity of demand (always explain PED concept whenever there are prices involved).

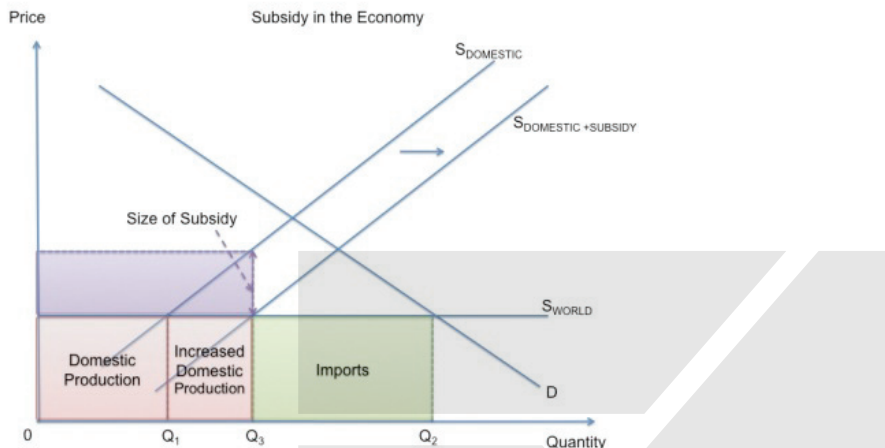


In the above figure, DD is the domestic demand and WS is the perfectly elastic world supply. At the world price of  $P_w$  the imports are  $Q_1$ . When the government imposed the quota of  $Q_2$  quantity, the supply becomes perfectly inelastic causing the price to rise to  $P_2$ . This will benefit the foreign exporter but will affect the domestic consumers will pay higher price causing their consumer surplus to fall by the area  $P_2ABP_w$ .

2. **Voluntary export restraints (VER)** are very likely quotas, but the two countries involved agree on the restriction. It might seem odd for the exporter to agree on the quota, but as discussed above, the revenue per unit sold increases. Until recently, Japan and the EU has such agreement for the importing of Japanese cars.
3. **Subsidies:** A subsidy is a grant paid by the government to a firm producing a certain good. It may be paid to promote the production of the good in question or, in this context, as a helping hand to domestic producers to help them compete with the foreign competition. A good example is the agriculture industry. The EU gives

## International trade

farmers across Europe billions of pounds. The USA also subsidizes their farmers (a sort of retaliation).



4. **Exchange controls:** Some countries still control the amount of foreign currency that is available so that the amount of imports that can be bought (with foreign currency) is reduced. In this situation all the commercial currency dealings are not allowed and the exporters are also required to surrender their foreign currency receipts and are paid in terms of domestic currency.
5. **Embargo:** It is the extreme form of quota restriction in which imports are banned mainly due to political reasons. For example Pakistan does not trade with Israel. UN imposed sanctions on trade with Iraq due to Iraqi attack on Kuwait.
6. **Other bureaucratic restrictions:** Countries often impose certain safety standard that are unnecessary, but put off foreign imports. In the past there have also been some ridiculous times - wasting regulations. There are many examples of countries that will only allow certain goods to be imported through a certain port or check point. There have also been examples of countries that take weeks to process the administration involved in admitting an import.

### Why protectionism is felt to be justified?

1. **Revenue rising:** Tariff has been used throughout history to provide the state with revenue. This is one of the motives, not the main one, of the European Union common external tariff.
2. **Protect domestic jobs:** During the great depression of the 1930s most countries resorted to increase protection of home industries in an attempt to maintain

## International trade

employment at home by diverting expenditure from foreign to domestic products. The idea is that money spent on home produced goods and services create employment at home while that spent on imports does not. But the imports of one country are the exports of another and restrictions on imports create unemployment in other countries. They are very likely to **retaliate** by protecting their own industries. There is the risk of a trade war occurring with barriers to trade becoming higher and higher and more widespread. As world exports decline all trading nations must, to some extent, be worse off.

3. **Prevention of dumping argument:** A strong argument can be advanced to protect industries against dumping. Foreign producers may be selling their products at less than cost in order to get rid of surplus stock and to gain a monopoly position in the export market by driving out domestic producers.
4. **The infant industry argument:** This is a rather old fashioned argument but countries do still protect infant industries if they can get away with it. If a country is planning to start up a new, but major, industry that is already established in many countries, then they will start at a huge disadvantage. It will take time for them to operate on a large scale and reduce their costs. This is recognized by the WTO (world trade organization) and some protection is allowed as these new industries find their feet.

In these developing countries the industrial infrastructure is usually not developed and the firms are usually operating on a small scale with a relatively higher cost of production and are unable to compete from fully developed large-scale producers abroad who operate at much lower costs. It is therefore, necessary to protect these sunrise industries from foreign competition until they get enough time to reduce their costs to be able to compete from the foreign producers. This protection is provided through imposition of tariffs on the imports of the goods produced by these infant industries. But the problem is that once these industries used to of protected economy in which they sell their goods at higher prices, they want to continue to enjoy heavy profits in the domestic economy indefinitely. They never grow up and reduce their costs as intended by the government and it becomes very difficult for the government to remove the tariffs.

In developed countries there are some industries which face a fall in their demand in the international market, may be because of the introduction of better or cheaper substitutes by the newly industrialized countries. It is difficult to close down these industries and divert the resources in the production of the other goods because most of the resources are not occupationally mobile, so these

## International trade

industries need to be protected by imposing tariffs to stop the imports of the goods produced by these industries. This will help these industries to make improvements in their products, explore more markets and regain their competitiveness in the international market.

5. **The strategic industry argument:** Many countries continue to produce goods, even though they could buy more cheaply from abroad. To give up production of these goods would be unwise for reasons of national security. Industries such as steel, agriculture, air-craft and ammunition manufacturing, ship-building power and so on are regarded as strategic industries which are essential for the survival of the country. It is regarded most desirable, therefore to protect these industries in the domestic economy so as to reduce a nation's dependence on foreign supplies of such strategic materials.
6. **To eliminate the Balance of Payments (BOP) deficit:** A BOP account is the record of the receipts and payments of a country with the rest of the world. If the payments are more than receipts the in the BOP is in deficit. If a country is persistently facing a deficit in the BOP, then it may finally impose restrictions such as tariffs and quotas to reduce imports into country. This will reduce the payment made and will help in eliminating the deficit in the BOP. Providing subsidies to domestic producers is another measure to increase exports causing the receipts to increase thereby reducing or eliminating the BOP deficit.
7. **Avoiding the risks of overspecialization:** A number of industries may be protected to ensure a diversified industrial structure.

### TERMS OF TRADE

**Terms of trade:** Terms of trade are defined as the quantity of domestic goods that must be exported to get a unit of imported goods. They reflect the opportunity cost of imports measured in terms of exports. It is also defined as the rate at which a country exchanges its exports for imports. In real world goods are not exchanged with the goods rather they have prices attached to them, so terms of trade is measured as the ratio of the index of export prices to the index of import prices expressed in percentage form.

$$\text{TOT} = (\text{Index of export prices} / \text{index of import prices}) \times 100$$

Index of export and import prices are the weighted averages of the major exports and imports of the country. The index is taken as 100 in some base year which is a relatively stable year. In the next if the terms of trade index increase to 105, it means that he

## International trade

relative export price has increased by 5% which is a favorable movement and it means that for every unit of the goods exported, the country will obtain a larger quantity of imports.

Terms of trade increases when export prices rise, import prices fall, both export prices and import prices rise but export rise faster, and both export and import prices fall but import prices fall faster. In all these four cases the relative export price rises.

If the terms of trade index number increases it is described as a favorable movement (a unit of exports will buy more imports).

If the terms of trade index number decrease as a worsening or deterioration of terms of trade (a unit of exports will buy fewer imports).

This terminology is somewhat ambiguous as it is important to bear in mind that a favorable or worsening of countries terms of trade does not necessarily mean that anything is better or worse in terms of balance of payments situation. Both the price and the quantity of goods traded must be taken in to account when considering the balance of payment situation.

An increase or favorable change in the Terms of Trade index caused by an increase in the price of exports may bring about proportionately greater fall in the demand for exports leading to worsening of the balance of payments situation. Conversely, worsening of the terms of trade caused by a fall in export prices may lead to a proportionately greater increase in the demand for exports and an improvement in the balance of payments.

It is important to consider how responsive the quantity demanded is to changes in the price of exports and imports. **The price elasticity of demand is thus crucial.** For example, if the demand for exports is elastic, then a fall in the prices of exports will lead to a more than proportionate increase in the demand for exports causing the export revenue to increase and the balance of payments position will improve even though the terms of trade worsened.

If the demand for exports and imports is elastic i.e. a change in prices brings a larger percentage change in quantity demand, then improvement in terms of trade represents that there is a rise in export price and fall in import prices. If demand is elastic, foreign residents will decrease the demand of that country where there is a favorable movement in terms of trade because the demand for goods is elastic, foreign residents have options in international market. Simultaneously there will be an increase in the demand for imported goods because they will appear cheaper than domestic goods, this rise in imports

## International trade

and fall in exports will worsen off balance of trade i.e. difference between exports revenue and imports expenditure.

On the other hand if the demand for exports and imports is inelastic i.e. a change in price brings a lesser percentage change in quantity demanded, then this rise in terms of trade will turn favorable for country in terms of balance of trade. A rise in exported goods prices won't have much effect on demand as those goods are a necessity for foreign countries. While for domestic residents where terms of trade has increased, imported goods will be now cheaper as they will have to pay for almost the same quantity a lesser amount or expenditure, thus, benefiting them.

